for some function of position $\mu(\boldsymbol{q})$. Plugging (7.33) and (7.34) into the PDE (7.32) gives

$$\begin{aligned}\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta} f_{2,1,[\![\mathrm{OABAO}]\!]}\right) &= -w(\boldsymbol{q},\boldsymbol{p})\rho_{\beta} - \mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}\nu\right), \\ &= -\frac{\beta}{24}\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\left[6\boldsymbol{p}^{T}\left(\nabla^{2}U\right)\nabla U - \frac{1}{3}\nabla^{3}U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right.\right. \\ &\quad \left.\left. -\frac{2}{\beta}\boldsymbol{p}^{T}\nabla_{q}\Delta_{q}U\right]\rho_{\beta}\right) - \bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\left[\boldsymbol{p}^{T}\nabla\nu + \mu\right]\rho_{\beta}\right),\end{aligned}$$

and hence

$$\begin{aligned}f_{2,1,[\![\mathrm{OABAO}]\!]} = &-\frac{\beta}{24}\left(6\boldsymbol{p}^{T}\left(\nabla^{2}U\right)\nabla U - \frac{1}{3}\nabla^{3}U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p} - \frac{2}{\beta}\boldsymbol{p}^{T}\nabla_{q}\Delta_{q}U\right) \\ &-\boldsymbol{p}^{T}\nabla\nu - \mu.\end{aligned}$$

At this point, we might consider abandoning the prospect of determining the leading order correction altogether, as we now have two arbitrary functions of position $\nu(\boldsymbol{q})$ and $\mu(\boldsymbol{q})$ creeping into our solution. However, the form of $\nu(\boldsymbol{q})$ is determined by considering the next order PDE at ε^2:

$$\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta} f_{2,2,[\![\mathrm{OABAO}]\!]}\right) = -\mathcal{L}_{H}^{\dagger}\left(\rho_{\beta} f_{2,1,[\![\mathrm{OABAO}]\!]}\right). \tag{7.35}$$

We are spared solving this equation by virtue of a solvability condition. Integrating the left hand side of (7.35) with respect to the momenta, we find

$$\int_{\mathcal{P}} \bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta} f_{2,2,[\![\mathrm{OABAO}]\!]}\right) \mathrm{d}\omega_{p} = 0,$$

by simple computation. Hence we also require the average of the right-hand side of (7.35) to be zero with respect to the momentum, or

$$\int_{\mathcal{P}} \mathcal{L}_{H}^{\dagger}\left(\rho_{\beta} f_{2,1,[\![\mathrm{OABAO}]\!]}\right) \mathrm{d}\omega_{p} = \bar{\pi}\mathcal{L}_{H}^{\dagger} f_{2,1,[\![\mathrm{OABAO}]\!]} = 0, \tag{7.36}$$

where $\bar{\pi}$ is the averaging operator

$$\left(\bar{\pi} f\right)(\boldsymbol{q}) = \int_{\mathcal{P}} f(\boldsymbol{q},\boldsymbol{p})\rho_{\beta}\mathrm{d}\omega_{p}.$$

Hence by rewriting the correction function as

$$f_{2,1,[\![\mathrm{OABAO}]\!]} = \frac{\beta}{24}\left(3\mathcal{L}_{\mathrm{A}}^{\dagger}|\nabla U|^2 - \frac{1}{3}\left(\mathcal{L}_{\mathrm{A}}^{\dagger}\right)^3 U - \frac{2}{\beta}\mathcal{L}_{\mathrm{A}}^{\dagger}\Delta_q U\right) + \mathcal{L}_{\mathrm{A}}^{\dagger}\nu - \mu,$$

we have

$$\bar{\pi}\mathcal{L}_H^{\dagger} f_{2,1,[\![\mathrm{OABAO}]\!]} = \frac{\beta}{24}\left(3\mathcal{M}|\nabla U|^2 - \frac{1}{3}\mathcal{N}U - \frac{2}{\beta}\mathcal{M}\Delta_q U\right) + \mathcal{M}\nu,$$

where we abbreviate

$$\mathcal{M}\psi(\boldsymbol{q}) = \bar{\pi}\mathcal{L}_H^{\dagger}\mathcal{L}_{\mathrm{A}}^{\dagger}\psi(\boldsymbol{q}) = \bar{\pi}\left((\mathcal{L}_{\mathrm{A}}^{\dagger})^2 + \mathcal{L}_{\mathrm{B}}^{\dagger}\mathcal{L}_{\mathrm{A}}^{\dagger}\right)\psi(\boldsymbol{q}) = \frac{1}{\beta}\Delta_q\psi(\boldsymbol{q}) - \nabla U\cdot\nabla_q\psi(q),$$

$$\mathcal{N}\psi(\boldsymbol{q}) = \bar{\pi}\mathcal{L}_H^{\dagger}(\mathcal{L}_{\mathrm{A}}^{\dagger})^3\psi(\boldsymbol{q}) = \bar{\pi}\left((\mathcal{L}_{\mathrm{A}}^{\dagger})^4 + \mathcal{L}_{\mathrm{B}}^{\dagger}(\mathcal{L}_{\mathrm{A}}^{\dagger})^3\right)\psi(\boldsymbol{q}) = \frac{3}{\beta}\mathcal{M}\left(\Delta_q\psi(\boldsymbol{q})\right).$$

Thus we have

$$\bar{\pi}\mathcal{L}_H^{\dagger}\rho_\beta f_{2,1,[\![\mathrm{OABAO}]\!]} = \frac{\beta}{8}\mathcal{M}\left(|\nabla U|^2 - \frac{1}{\beta}\Delta_q U\right) + \mathcal{M}\nu,$$

and hence by (7.36) we finally obtain

$$\nu(\boldsymbol{q}) = \frac{1}{8}\left(\Delta_q U - \beta|\nabla U|^2\right),$$

where it is easily verified that $\langle\nu\rangle = 0$. This gives us our leading order solution

$$f_{2,0,[\![\mathrm{OABAO}]\!]} = \frac{1}{8}\left(\Delta_q U - \beta|\nabla U|^2\right).$$

Applying the same procedure for the $[\![\mathrm{OBABO}]\!]$ scheme yields

$$f_{2,0,[\![\mathrm{OBABO}]\!]} = \frac{1}{8}\left(\beta|\nabla U|^2 - 2\Delta_q U + c\right),$$

where $c = \langle\Delta U\rangle$ is a constant chosen to ensure zero average. We can relate these schemes to their conjugate schemes with centralized OU update via Lemma 7.1 and Eq. (7.30), to give

$$f_{2,0,[\![\mathrm{ABOBA}]\!]} = \frac{1}{8}\left(\beta\boldsymbol{p}^T\nabla^2 U(\boldsymbol{q})\boldsymbol{p} - 2\Delta_q U + c\right),$$

$$f_{2,0,[\![\mathrm{BAOAB}]\!]} = \frac{1}{8}\left(\Delta U - \beta\boldsymbol{p}^T\nabla^2 U(\boldsymbol{q})\boldsymbol{p}\right).$$